Quiz 1 (*Aug 17, Marks: 10, Duration: 50 mins*) **COL703/7203, Jul-Nov 2026**

Name: **Entry No.:**

1. **[2 marks]** Suppose that ϕ and ψ are formulas such that $\phi \vDash \psi$. Show that if ϕ and ψ have no variable in common then either ϕ is unsatisfiable or ψ is valid.

Space for rough work

2. **[2 marks]** Give a natural deduction proof of the law of excluded middle using basic proof rules.

Space for rough work

3. **[1 marks]** Is it true that we can transform any (natural deduction) proof of $\phi_1, \phi_2, \ldots, \phi_n \vdash \psi$ into a proof of the theorem $\vdash \phi_1 \to (\phi_2 \to (\phi_3 \to (\ldots \to (\phi_n \to \psi)\ldots)))$? If not, explain why. If yes, sketch the transformation.

Space for rough work

4. **[5 marks]** Recall that if we have clauses $(p \vee q)$ and $(r \vee \neg q)$, then we can apply resolution to get the clause $(p \vee r)$. Does this mean that the formula $(p \vee q) \wedge (r \vee \neg q)$ is equivalent to $(p \vee r)$? In particular:

   (a) **[2 marks]** Is $(p \vee q) \wedge (r \vee \neg q) \vdash (p \vee r)$ valid? If so, give a natural deduction proof for this. If not, justify your answer using a suitable valuation.

Space for rough work

(b) **[2 marks]** Is $(p \vee r) \vdash (p \vee q) \wedge (r \vee \neg q)$ valid? If so, give a natural deduction proof for this. If not, justify your answer using a suitable valuation.

Space for rough work

(c) **[1 marks]** Two formulas are said to be *equisatisfiable* if the first formula is satisfiable whenever the second is and vice versa; in other words, either both formulas are satisfiable or neither is. The truth values of two equisatisfiable formulas may nevertheless disagree for a particular assignment of variables.

Are the two formulas $((p \vee q) \wedge (r \vee \neg q)$, and $(p \vee r))$ equisatisfiable? Why or why not?

Space for rough work